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CE F231 Fluid Mechanics

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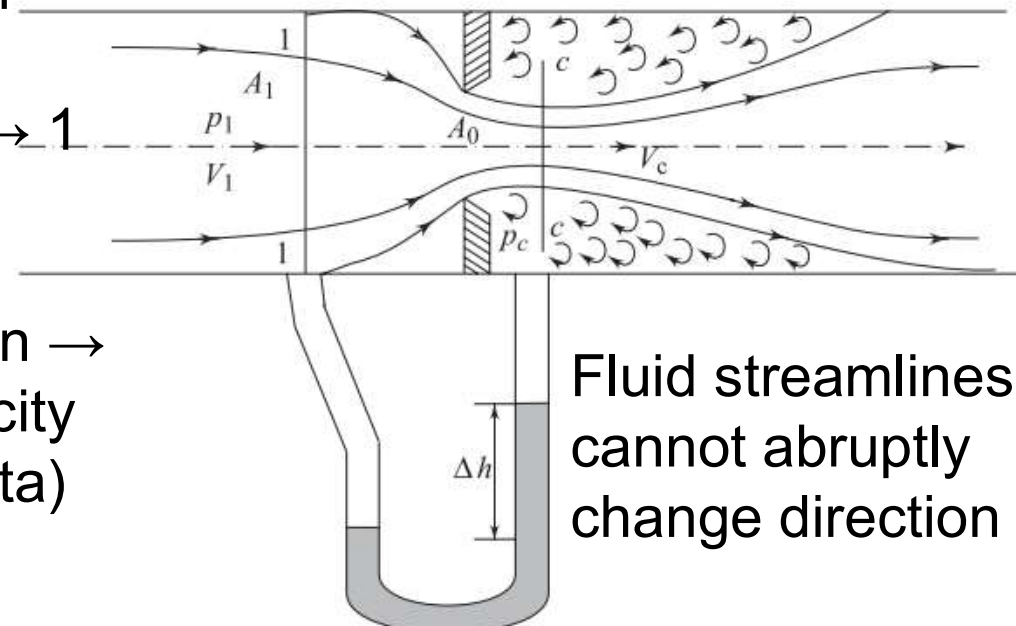
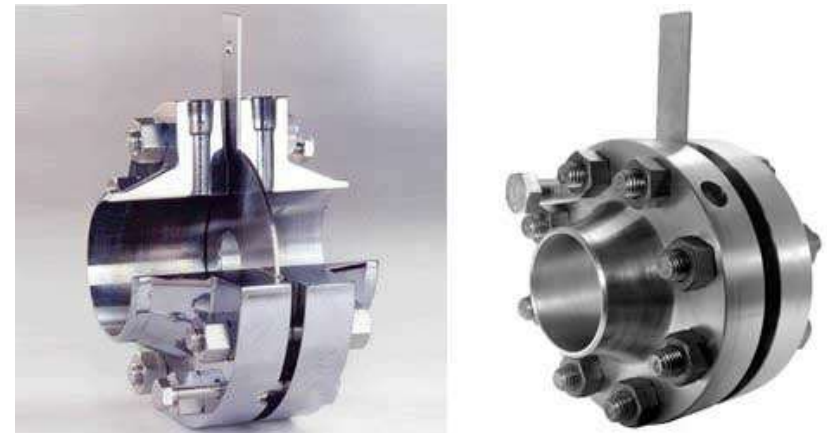
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Lecture 24: Fluid Dynamics

November 9, 2022

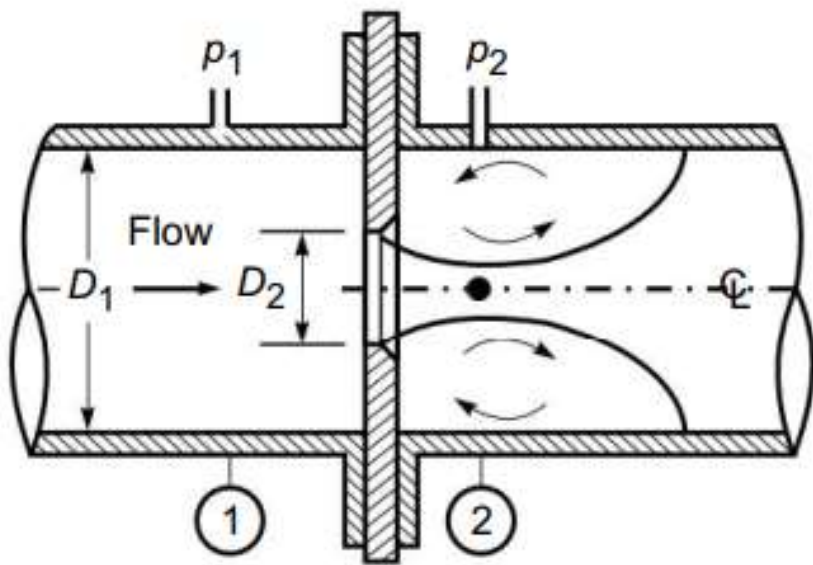
Recap: Orifice meter

- Used for measuring volume flow rate of fluid through pipe
- Simpler and cheaper than venturi meter
- Thin circular plate with sharp edged concentric circular hole
- Orifice diameter is 0.5 x pipe diameter
- Upstream pressure tapping location → 1 to 2 x pipe diameter
- Downstream pressure tapping location → 0.5 x orifice diameter (maximum velocity or minimum pressure at vena contracta)



Recap: Orifice meter

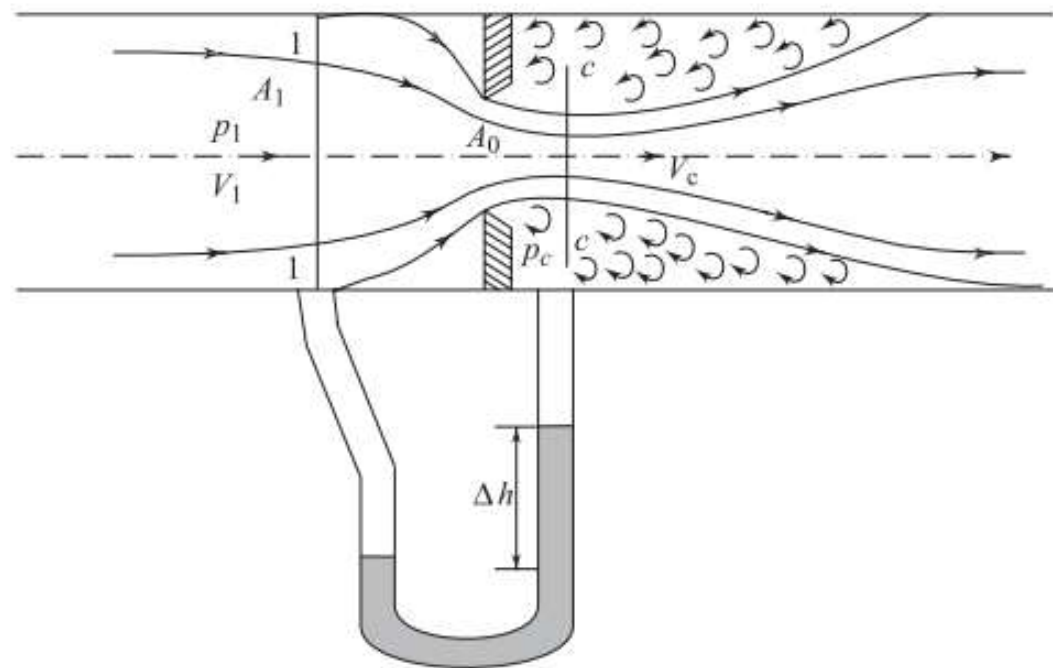
An orifice meter with orifice diameter 15 cm is inserted in a pipe of 30 cm diameter. The pressure gauges fitted upstream and downstream of the orifice meter give readings of 14.715 N/cm^2 and 9.81 N/cm^2 respectively. Find the rate of flow of water through the pipe in litres/s. Take $C_d = 0.6$. [Ans. 108.434 lit/s]
 $C_v = 0.98$



$$Q = C_d A_2 \frac{\sqrt{2g\Delta H}}{\sqrt{1 - C_c^2 (D_2/D_1)^4}} \quad (13.7)$$

where $\Delta H = \frac{(p_1 - p_2)}{\gamma}$ = difference in pressure heads upstream and downstream of the orifice plate.

Recap: Orifice meter

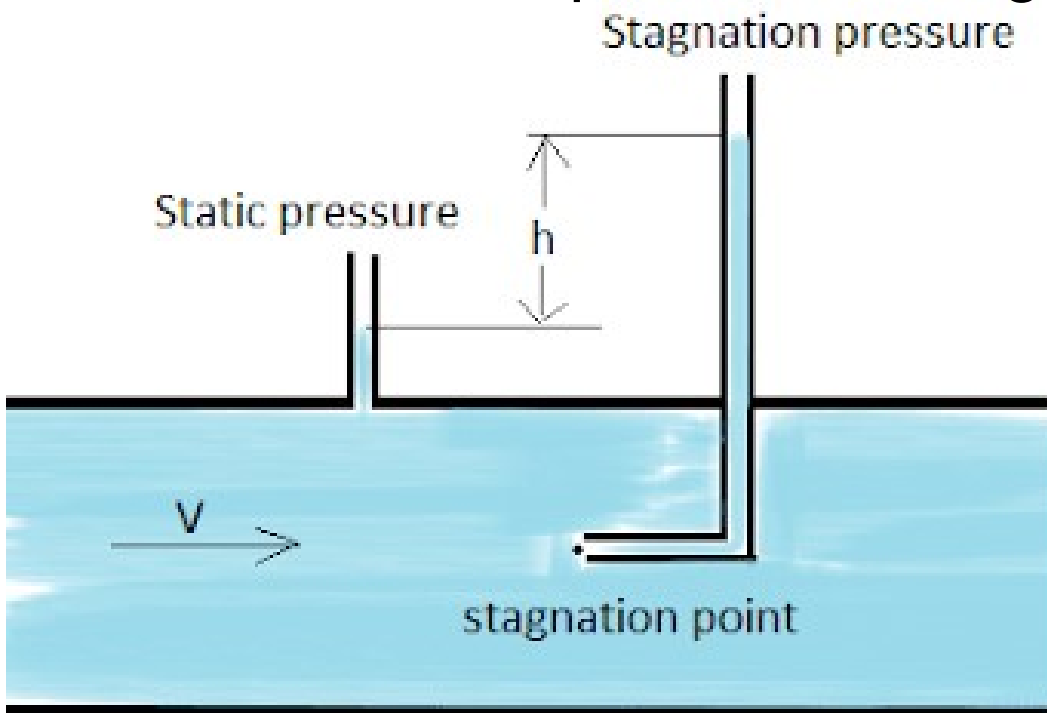


$$Q = C_c A_0 C_v \sqrt{\frac{2g(\rho_m / \rho - 1)}{(1 - C_c^2 A_0^2 / A_1^2)}} \Delta h$$

where Δh is the difference in liquid levels in the manometer and ρ_m is the density of the manometric liquid.

Pitot tube

- Device used for measuring velocity at a point in a flowing fluid
- If velocity at a point in a flow becomes 0, then kinetic energy is converted into pressure energy



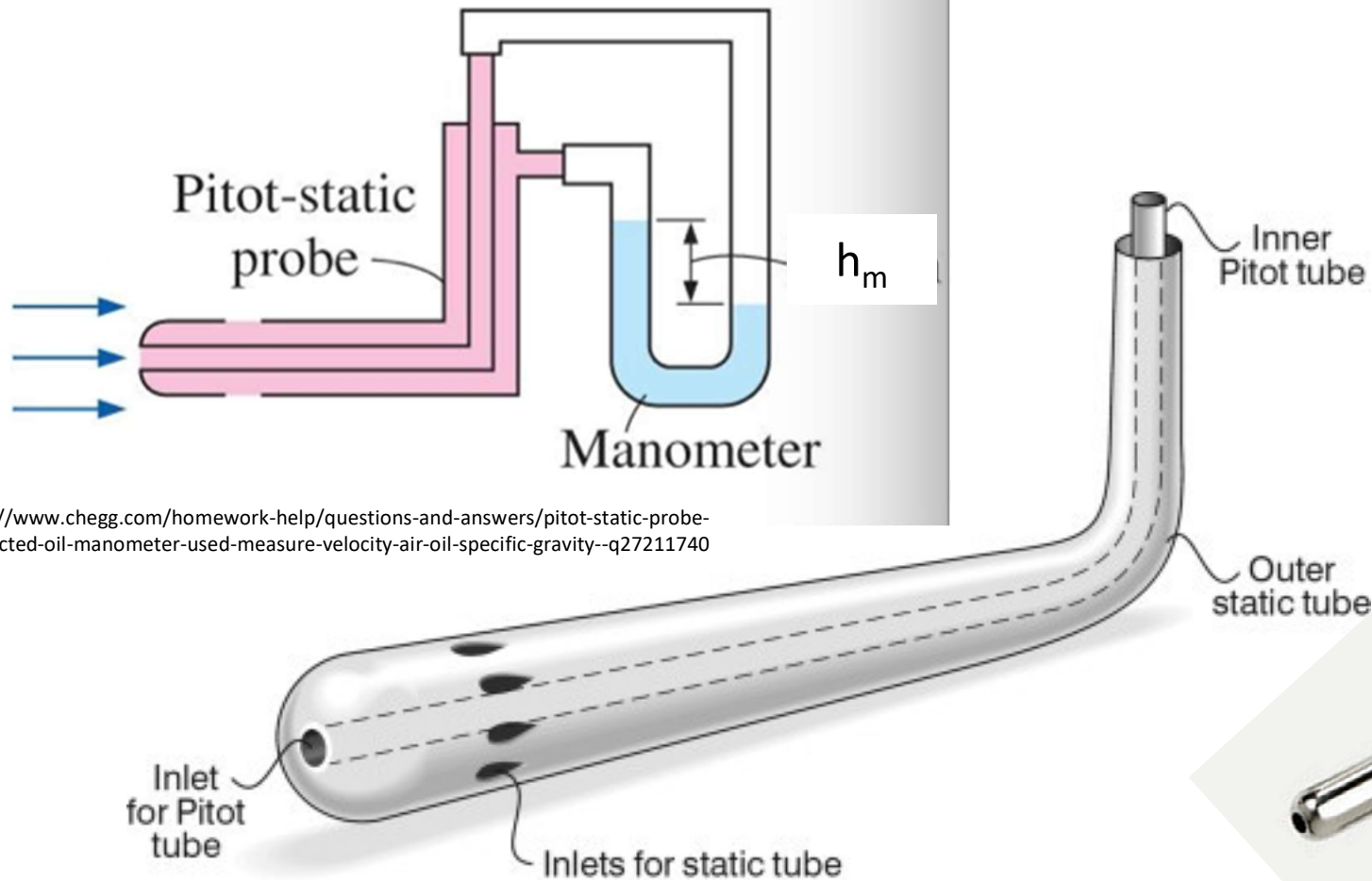
Simple pitot tube

Simple pitot tube
(Single tube
Inner tube only –
stagnation pressure)

Pitot static tube
(Concentric tubes
Inner tube – stagnation pressure;
outer tube – static pressure)

Stagnation pressure = static + dynamic

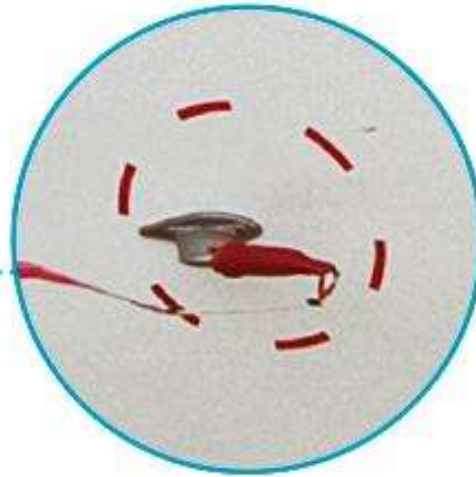
Pitot static tube



<https://www.chegg.com/homework-help/questions-and-answers/pitot-static-probe-connected-oil-manometer-used-measure-velocity-air-oil-specific-gravity-q27211740>

Inner tube: Stagnation pressure; Outer tube: Static Pressure

Pitot tube



<https://safetyfirst.airbus.com/pitot-probe-performance-covered-on-the-ground/>

<https://www.keyshone.com/velocity-measurement-pitot-static-tube-subsonic-supersonic-regime/>

<https://www.defencematrix.in/pitottube>

Pitot tube



https://www.123rf.com/photo_81341944_the-air-pressure-pitot-tube-31-mig-on-the-codification-of-nato-foxhound-fox-hound-.html

<https://www.defencematrix.in/pitottube>

Pitot tube



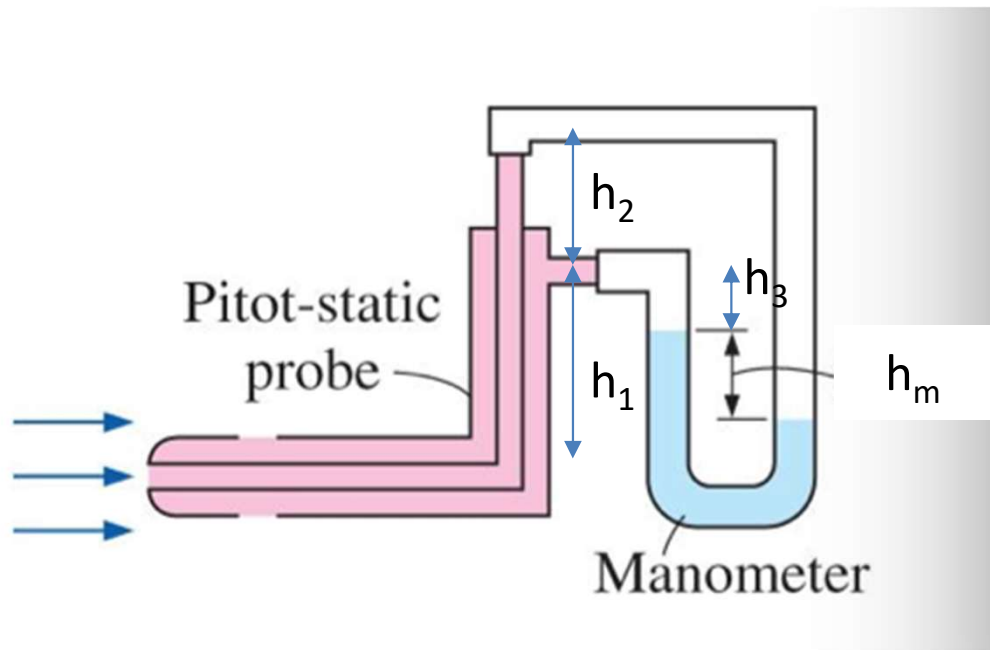
The pressure difference measured by the two tappings of a pitot-static tube, one tapping pointing upstream and other perpendicular to the flow, placed in the centre of a pipe line of diameter 40 cm is 10 cm of water. The mean velocity in the pipe is 0.75 times the central velocity. Find the discharge through the pipe. Take co-efficient of pitot-tube as 0.98.

[Ans. 0.1293 m³/s]

Pitot tube



A sub-marine moves horizontally in sea and has its axis 20 m below the surface of water. A pitot-static tube placed in front of sub-marine and along its axis, is connected to the two limbs of a *U*-tube containing mercury. The difference of mercury level is found to be 20 cm. Find the speed of sub-marine. Take sp. gr. of mercury 13.6 and of sea-water 1.026. [Ans. 24.958 km/hr.]



Inner tube: Stagnation pressure; Outer tube: Static Pressure